

Motor Datasheet: WxF-70x24GT

Machine Configuration

Parameter	Value	Unit
Machine Outer Diameter	68.0	mm
Machine Length / Thickness	23.7	mm
Voltage	48.0	V
Operating Temperature	-40 to 85	°C

Mass Properties

Parameter	Value	Unit
Machine Mass (active components only)	377.0	g
Rotor Mass	101.6	g
Stator Mass	275.4	g
Rotor Structure Included	No	-
Rotor Moment of Inertia	689.9	g·cm ²

Performance Metrics

Parameter	Value	Unit
Torque (continuous)	2.60	Nm
Speed	2500	rpm
Losses (continuous)	62.7	W
Supply Current (RMS)	13.32	A
Phase Resistance (per phase)	96.7	mΩ
Terminal Inductance ($L_{tt} = L_d + L_q$)	531.8	μH
D-axis Inductance (L_d)	272.6	μH
K_e (peak, phase)	9.55	mV/rpm
K_e (peak, line-to-line)	16.53	mV/rpm
K_t (peak)	136.7	mNm/A
Motor Constant (K_m)	362.1	mNm/√W